



End-of-Unit Test

Name Date Class

Standards: 9.3.5.1, 9.3.5.3, 9.3.5.5, 9.3.5.7, 9.3.5.8 · Directions: Show all work. Work must be turned in for full credit.

SECTION A — VARIABLES & TRANSLATING

1. [1 pt] Write an algebraic expression: "A number divided by 6." A. $6x$ B. $x / 6$ C. $x - 6$ D. $x + 6$

2. [1 pt] Write an algebraic expression: "The sum of a number and 9." A. $9x$ B. $x - 9$ C. $x + 9$ D. $x / 9$

3. [1 pt] Write an algebraic expression: "Three times a number, increased by 8." A. $3x + 8$ B. $3(x + 8)$
C. $x + 24$ D. $8x + 3$

4. [1 pt] Write an algebraic expression: "5 less than the product of 4 and a number." A. $5 - 4x$ B. $4x - 5$ C. $4(x - 5)$ D. $20x$

5. [1 pt] Write an algebraic expression: "The quotient of a number and 2, decreased by 10." A. $x / 2 - 10$ B. $2x - 10$ C. $(x - 10) / 2$ D. $x / (2 - 10)$

6. [1 pt] Write an expression: "Half of a number, added to 3 times the same number." A. $x/2 + 3x$ B. $3.5x$ C. $(x + 3x)/2$ D. $3x/2 + x$

SECTION B — ORDER OF OPERATIONS (GEMDAS)

7. [1 pt] Evaluate: $5 + 7 \times 2$ A. 24 B. 19 C. 14 D. 35

8. [1 pt] Evaluate: $(9 - 4) \times 3$ A. 3 B. 15 C. 27 D. 12

9. [1 pt] Evaluate: $18 - 6 \div 2 + 4$ A. 10 B. 4 C. 19 D. 8

10. [1 pt] Evaluate: $3 \times (6 + 2) - 4^2$ A. 8 B. -10 C. 32 D. 40

11. [1 pt] Evaluate: $12 - 8 \div 4 + 3$ A. 4 B. 1 C. 13 D. 7

12. [1 pt] Evaluate: $2(3 + 4) - 5$ A. 9 B. 14 C. 5 D. -9

SECTION C — EVALUATING EXPRESSIONS

13. [1 pt] Evaluate $m + 15$ when $m = 8$. A. 23 B. 7 C. 120 D. -7

14. [1 pt] Evaluate $4n - 7$ when $n = 5$. A. -3 B. 13 C. 27 D. 8

15. [1 pt] Evaluate $3x + 2y$ when $x = 4$ and $y = 6$. A. 14 B. 24 C. 34 D. 48

16. [1 pt] The perimeter of a square is $P = 4s$. Find P when $s = 9.5$. A. 13.5 units B. 38 units C. 90.25 units D. 9.5 units

SECTION D — SIMPLIFYING & EQUIVALENT EXPRESSIONS

17. [1 pt] How many terms are in $7x^2 - 4x + 2$? A. 2 B. 3 C. 4 D. 1

18. [1 pt] Simplify: $9x + 4x - 2x$ A. $7x$ B. $15x$ C. $11x$ D. $3x$

19. [1 pt] Simplify: $6y + 5 - 2y + 8$ A. $4y + 3$ B. $4y + 13$ C. $8y + 13$ D. $4y - 3$

20. [1 pt] Simplify: $5(2x - 3)$ A. $10x - 3$ B. $7x - 8$ C. $10x - 15$ D. $2x - 15$

21. [1 pt] Simplify: $3(x + 4) + 5x$ A. $8x + 12$ B. $8x + 4$ C. $15x + 12$ D. $3x + 20$

22. [1 pt] Are $2x + 3$ and $x + x + 3$ equivalent? A. Yes, because $2x = x + x$ B. No, they have different terms C. Only when $x = 0$ D. No, because of the constant

SECTION E — FACTORING & PROPERTIES

23. [1 pt] Factor the GCF: $6x + 12$ A. $6(x + 6)$ B. $6(x + 12)$ C. $6(x + 2)$ D. $3(2x + 4)$

24. [1 pt] Simplify: $-3(2x - 4) + 2x$ A. $-4x - 12$ B. $-8x + 12$ C. $-4x + 12$ D. $-6x + 12$

25. [1 pt] Factor the GCF: $8x^2 + 4x$ A. $4x(2x + 1)$ B. $2x(4x + 2)$ C. $8(x^2 + x)$ D. $4(2x^2 + x)$

SECTION F — REAL-WORLD APPLICATIONS

26. [1 pt] Mia earns \$12 per hour babysitting. Write an expression for her earnings after h hours. A. $12 + h$ B. $12h$ C. $h / 12$ D. $12 - h$

27. [1 pt] A jacket costs \$45 at Target. It's on sale for 20% off. What is the sale price? A. \$9 B. \$25 C. \$36 D. \$40

28. [1 pt] A shirt costs s dollars. Buy 2, get 1 free. What's the expression for 5 shirts? A. $4s$ B. $5s$ C. $3s$ D. $2s$

29. [1 pt] A rectangle has length $L = 2x$ and width $W = x + 3$. What is its perimeter? A. $6x + 6$ B. $3x + 3$ C. $6x + 3$ D. $4x + 6$

30. [1 pt] What is the coefficient of x in the expression $5x - 3y + 2$? A. 5 B. -3 C. 2 D. 3

SECTION G — SCIENTIFIC NOTATION

31. [1 pt] Write the number 0.00052 in scientific notation. A. 5.2×10^{-4} B. 52×10^{-5} C. 5.2×10^4 D. 0.52×10^{-3}

32. [1 pt] Write 6.4×10^5 in standard form. A. 640,000 B. 64,000 C. 6,400,000 D. 640 E. 64

Total points: 32 · End of assessment.